Timothy Bielefeld

Embedded Control Systems & Firmware Development
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Education

Purdue University West Lafayette, IN
Bachelor of Science in Electrical Engineering *Aug. 2024 – May 2028*

- GPA: 3.88; Concentration in Automatic Control; Dean's List/Semester Honors, 2024-2026

Experience

Electrical Engineering Intern Jun. 2026 – Aug. 2026
Sports Concepts Lab *Reston, VA*

- Developed non-blocking C++ firmware driving a 185 Hz linear resonant actuator through an I2C haptic driver, using auto-calibration to achieve 2G peak feedback.
- Designed KiCad schematics for an ESP32-C3 wearable, integrating battery protection and power management to sustain 4+ hours of active runtime.
- Built a React Native/TypeScript app to configure athlete wearables over BLE, supporting deployment across squads of 20–30 athletes.
- Engineered the app's connection handling to manage up to 7 simultaneous BLE links per session, working within Bluetooth's concurrent-connection limit.

Projects

Optical Heart Rate Sensor | *Analog Circuits, Op-Amps, FRA* Apr. 2026

- Designed a complete four-stage analog signal chain — infrared sensing, active filtering, amplification, and comparator-based detection — converting a raw PPG signal into a clean digital heartbeat pulse.
- Measured 54.87 V/V peak filter gain via frequency response analysis, validating the sensor's ability to isolate cardiac signal from noise.
- Synchronized a digital output and LED indicator with each detected heartbeat, demonstrating a full pipeline from analog front-end to usable digital signal.
- Diagnosed and corrected noise and drift issues across analog filter stages through iterative oscilloscope testing, improving pulse-detection reliability under real-world lighting conditions.

Smart HVAC System for Microclimate | *Arduino, C++, Feedback Control, PWM* May 2026

- Built a closed-loop thermal regulation system using an Arduino, motor driver, DC ventilation fan, and temperature sensor.
- Implemented discrete-loop C++ control to compensate for thermal dead time and minimize overshoot, stabilizing target temperature within ±0.5°C.
- Added a local LCD interface for real-time setpoint adjustment and live system-state diagnostics.

Audio Equalizer | *Op-Amps, RC Filters, LTspice* Dec. 2025

- Designed, simulated in LTspice, and built a 3-band audio equalizer using op-amp gain stages and RC filters.
- Validated hardware cutoff frequencies with an oscilloscope across a wide range of music inputs, maintaining response within ±2 dB from 100 Hz–10 kHz.

Cruise Control Algorithm | *MATLAB, Modeling, Simulation* May 2025

- Led, modeled, and implemented a MATLAB cruise control algorithm evaluating stability under varying vehicle and tire dynamics.
- Used noisy sensor data to stress-test controller performance, validating stability margins across multiple simulated driving conditions.

Technical Skills

Embedded & Hardware: Microcontrollers (ESP32, Arduino), PCB Design (KiCad), I2C, BLE, PWM, Circuit Design
Controls & Systems: PID Loop Tuning, Feedback Control, Frequency Response Analysis (FRA)
Programming: C/C++, Python (NumPy, SciPy), MATLAB, Java, TypeScript, HTML/CSS, Verilog
Tools: Oscilloscope, LTSpice, Git/GitHub, Expo, React Native, SQL, Excel